College of Science and technology
Computer Engineering
Year 3

Lab 5: Posts Manager with SQLite

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Mobile application systems and design
3/25/26

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Lab 5: Posts manager with SQLite.

Question 1: Which dependencies did you use and why? Explain why SQLite is necessary to store data locally.

Answer.

I used the following dependencies:

*1 **sqflite:** This is the main package used to implement SQLite in Flutter. It allows the app to create a database and perform CRUD operations such as update, delete, insert data.

*2 **path:** This package is used to define the correct storage location of the database file on the device.

SQLite is necessary because the application is designed to work offline. It allows data to be stored locally on the device without requiring an internet connection. This ensures that posts remain available even after the app is closed or restarted.

Question 2: How do you handle database exceptions?

Answer.

Database exceptions are handled using try-catch blocks around all database operations.

* If the database is not initialized, the app ensures it is created before performing operations.

* Errors during insert, update or delete are caught and handled without crashing the app.

* A SnackBar message is displayed to inform the user when an operation fails.

This approach ensures stability and prevents unexpected application cra-

shes.

- Handle database not initialized.

Before performing any operation, the app checks whether the database is initialized. If not initialized, the database is created using a helper class. This is done using a singleton pattern (DatabaseHelper instance) to ensure only one database instance exists.

This guarantees that all operations are executed on a valid database.

- Handle insert/update/delete errors.

Errors during CRUD operations are handled using exception handling: Insert errors prevent invalid or empty data from being saved. Update errors ensure that only existing records are modified. Delete errors ensure the record exists before deletion.

Incase of failure, the app catches the error, shows a user friendly message (SnackBar) and prevents app crashes.

- Handle invalid or corrupted data.

The app validates input before saving; this ensures fields like title and body are not empty. Also this prevents null values from being inserted. If corrupted data is detected, the operation is rejected, and the user is notified. This maintains data integrity in the database.

Question 3: Explain SQLite in Flutter.

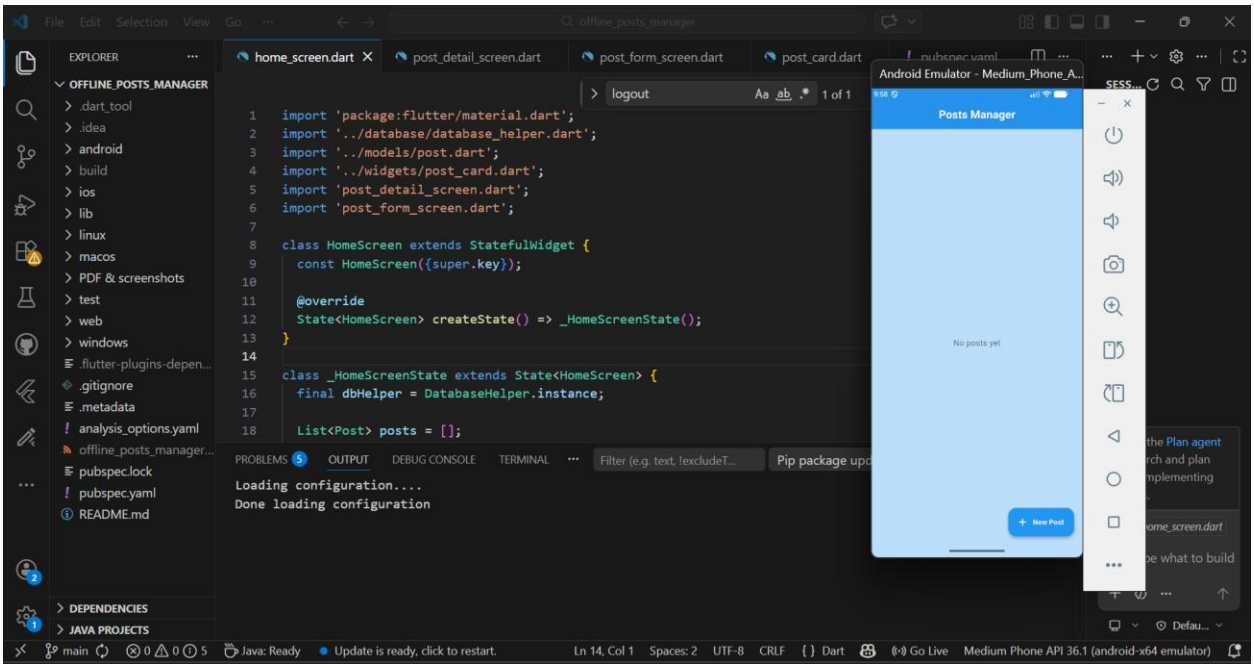
Answer: SQLite is a local database stored on the device. In Flutter, it is used through the sqflite package to store and manage structured data without requiring internet access.

- A database is a collection of data, while a table is a structure within the database that organizes data into rows and columns.
- CRUD operation includes Create (add post), Read (view posts), Update (edit post), Delete (remove post), which are the core operations of the app.
- Flutter uses asynchronous and await to perform database operations in the background, ensuring the UI remains responsive.

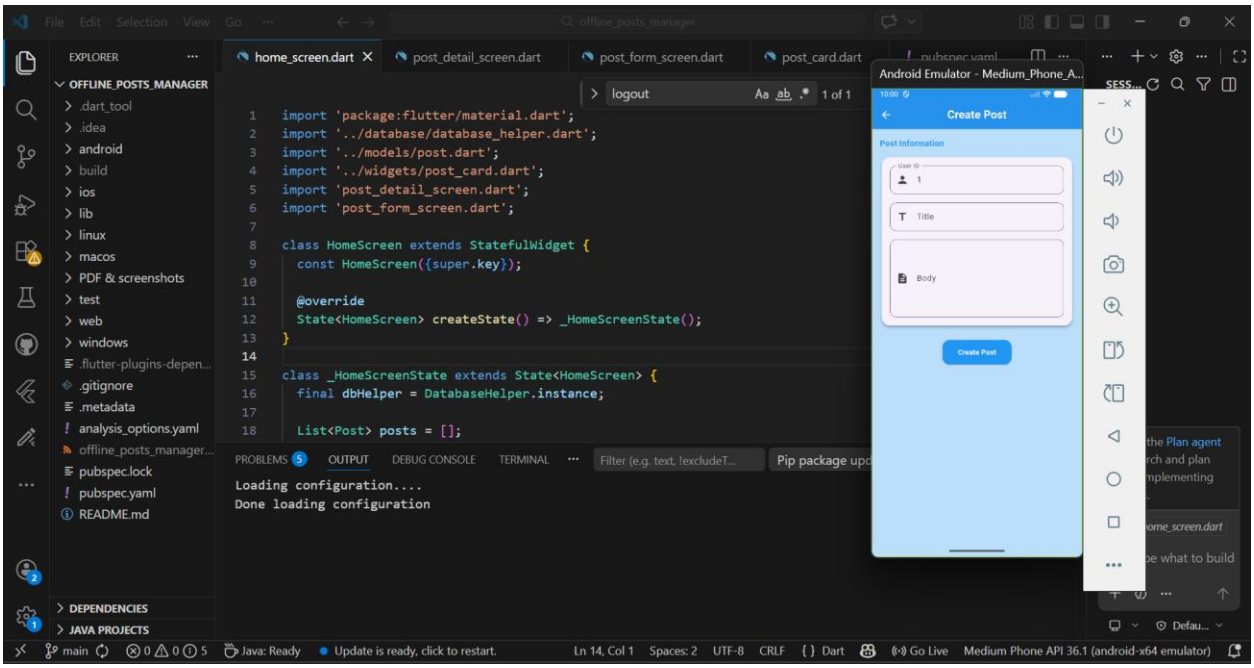
As conclusion: The Offline Posts Manager app demonstrates how Flutter can use SQLite for local data storage. By implementing CRUD operations and handling errors effectively, the app provides a reliable and fully functional offline system.

Screenshots of Offline Posts Manager with SQLite

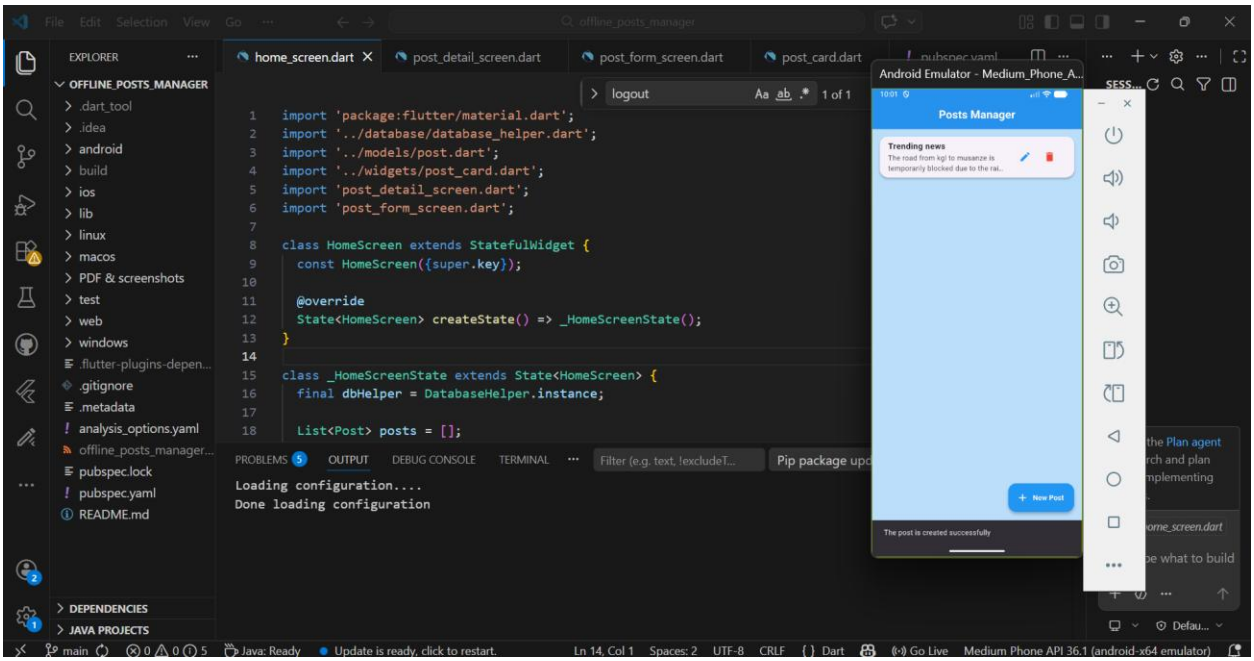
1. Home screen



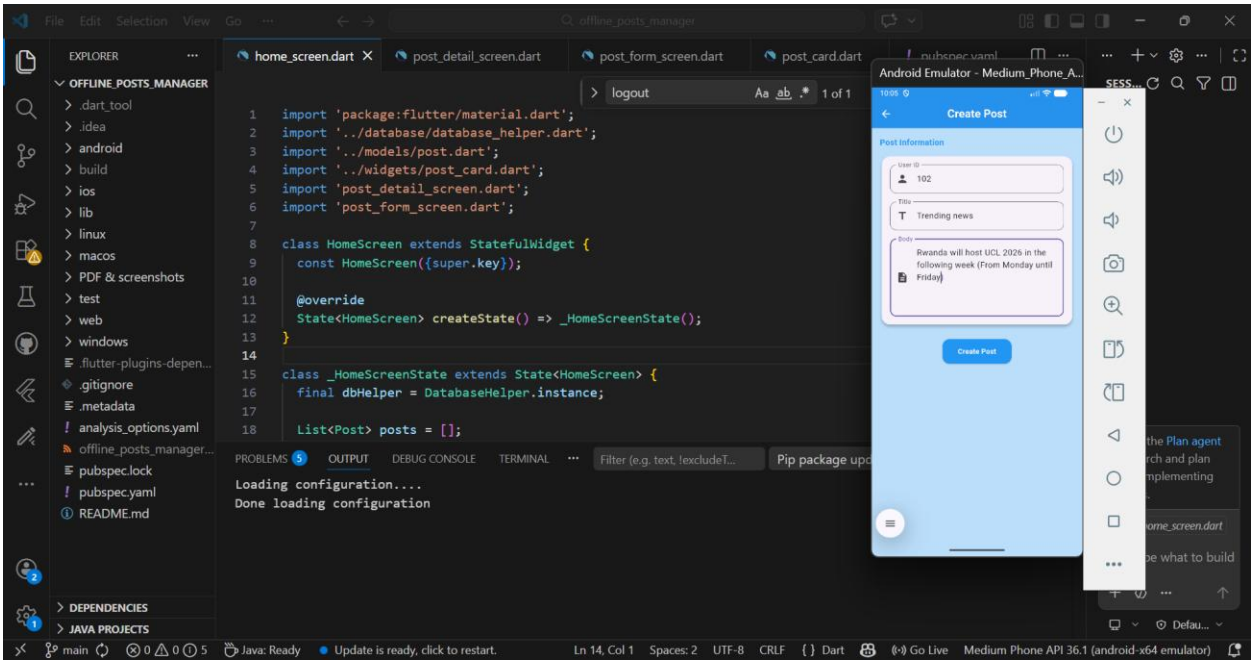
2. Postform screen

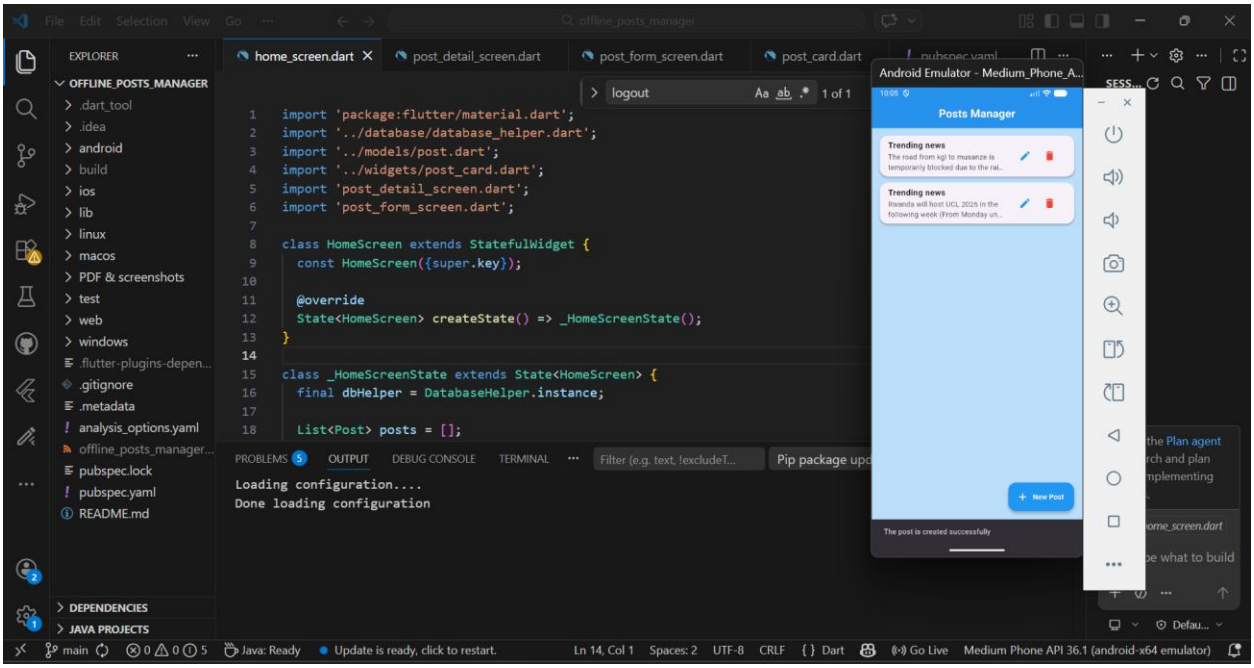


3. Creation of first post

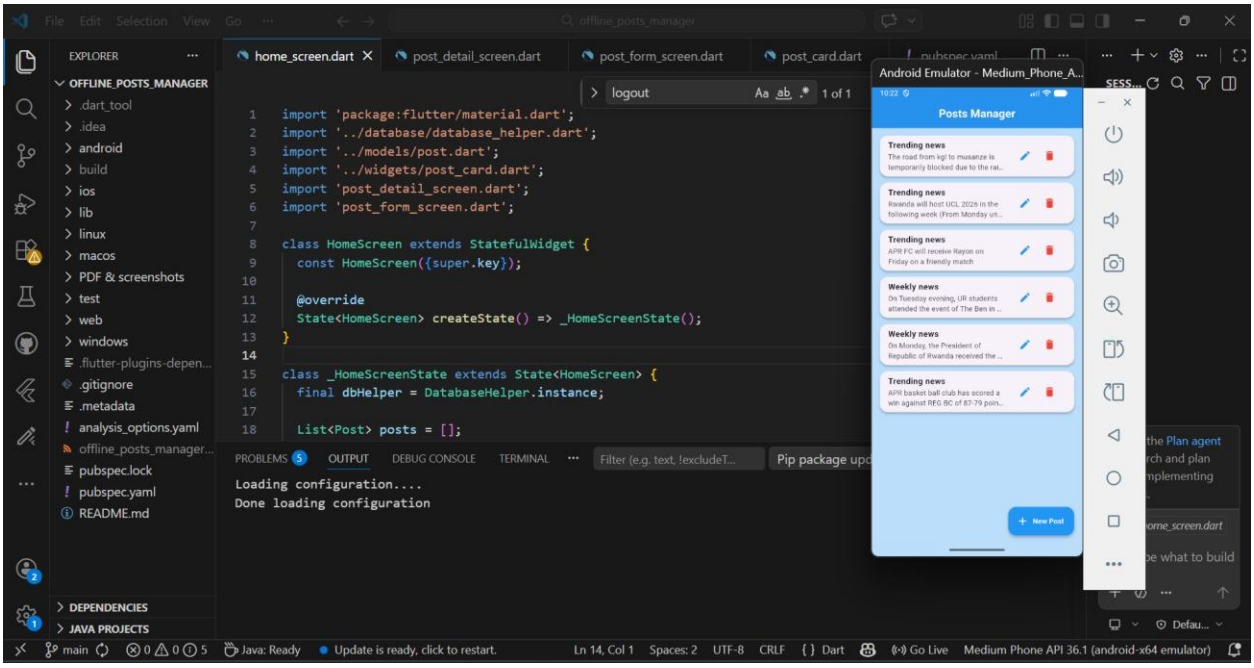


4. Creation of second post

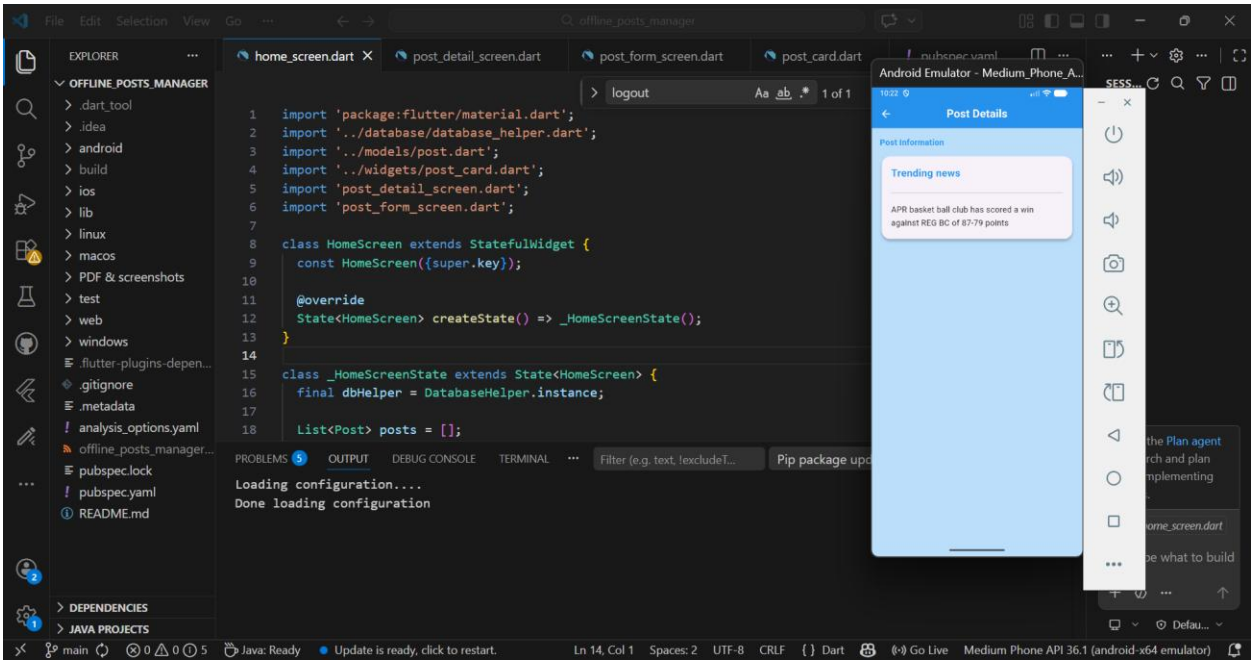


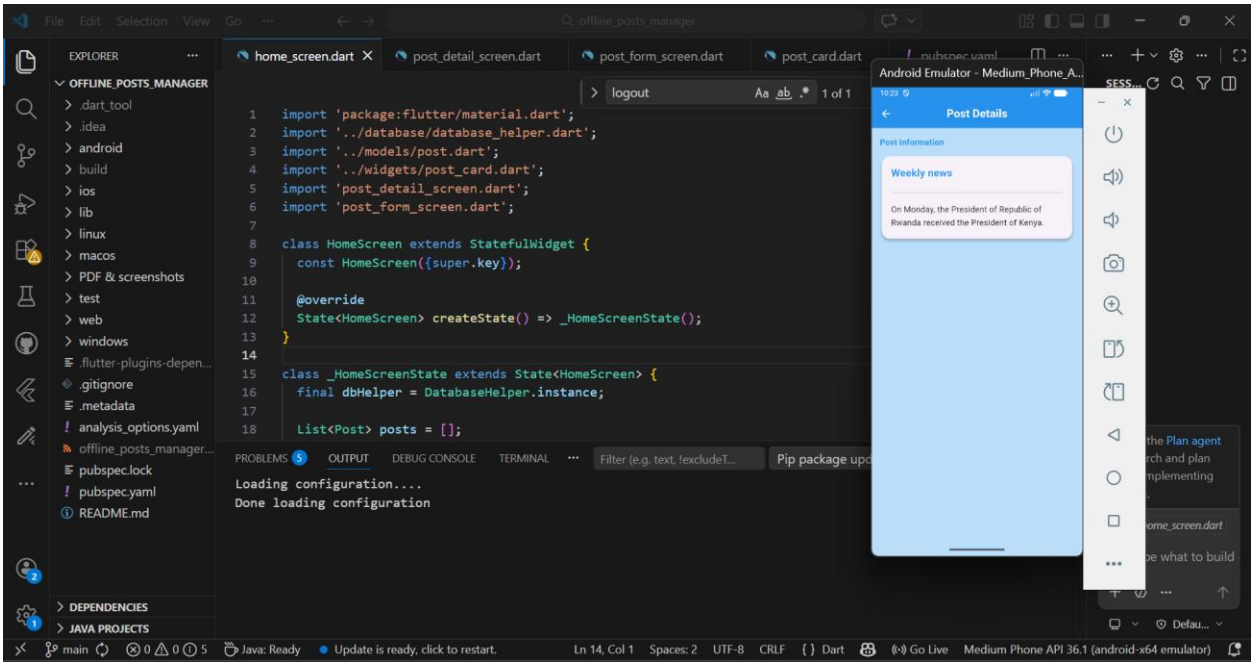


5. Creation of other posts

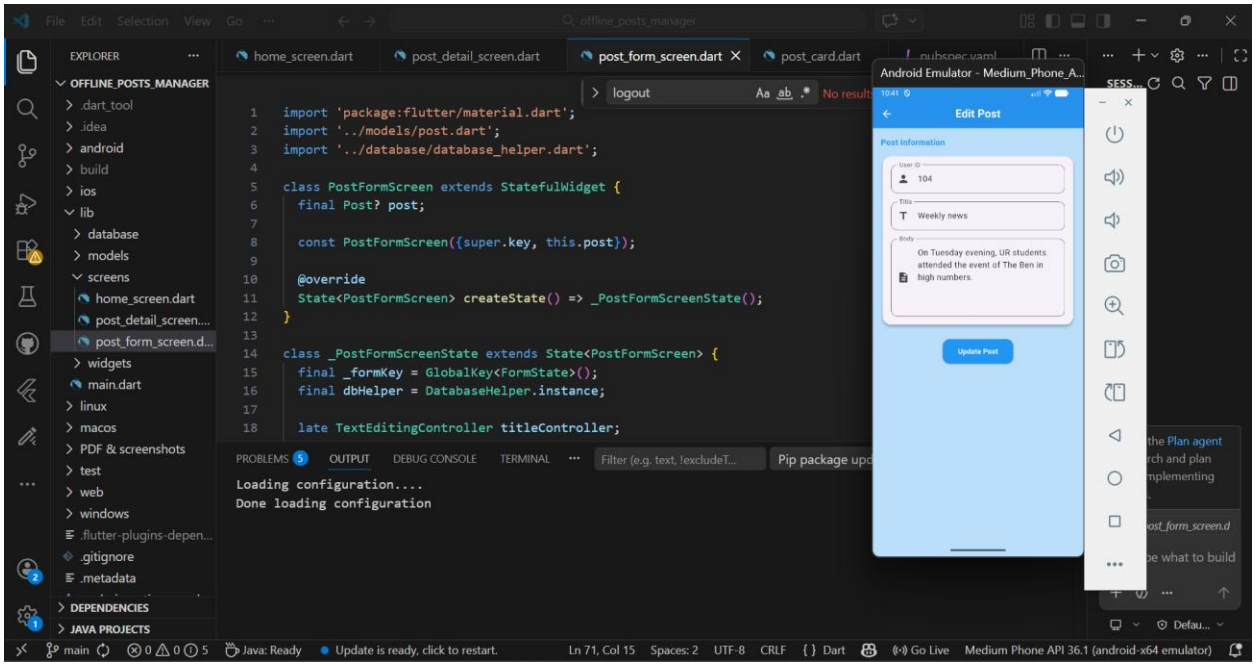


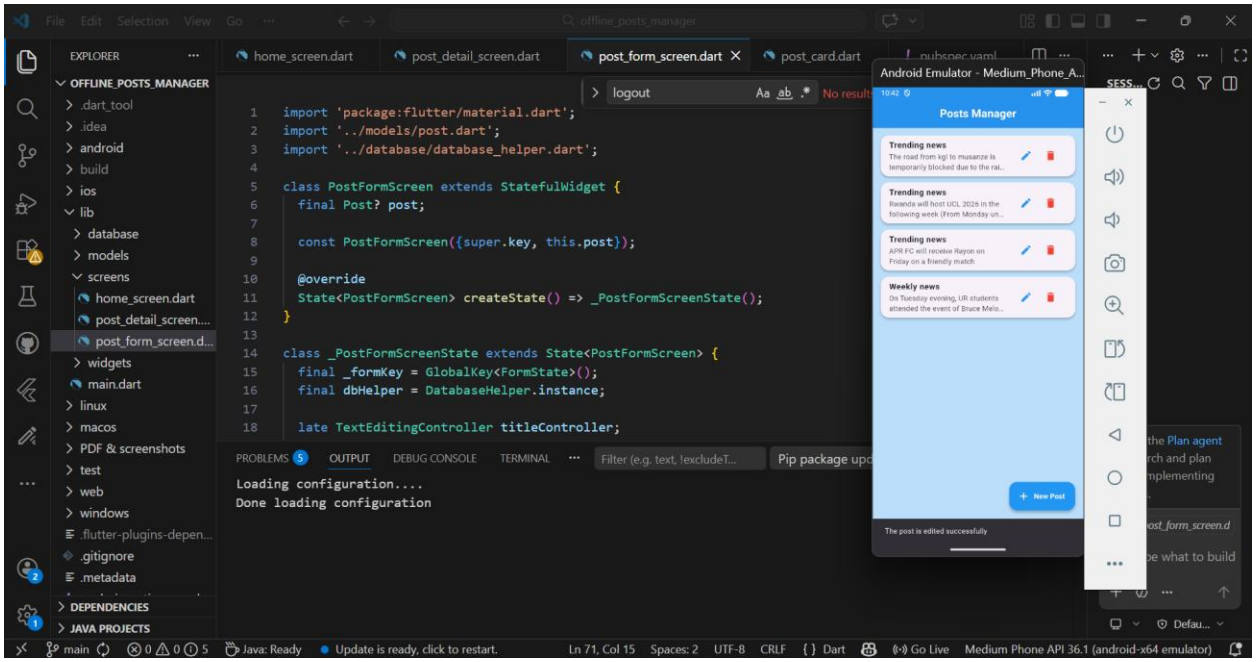
6. Post details screen



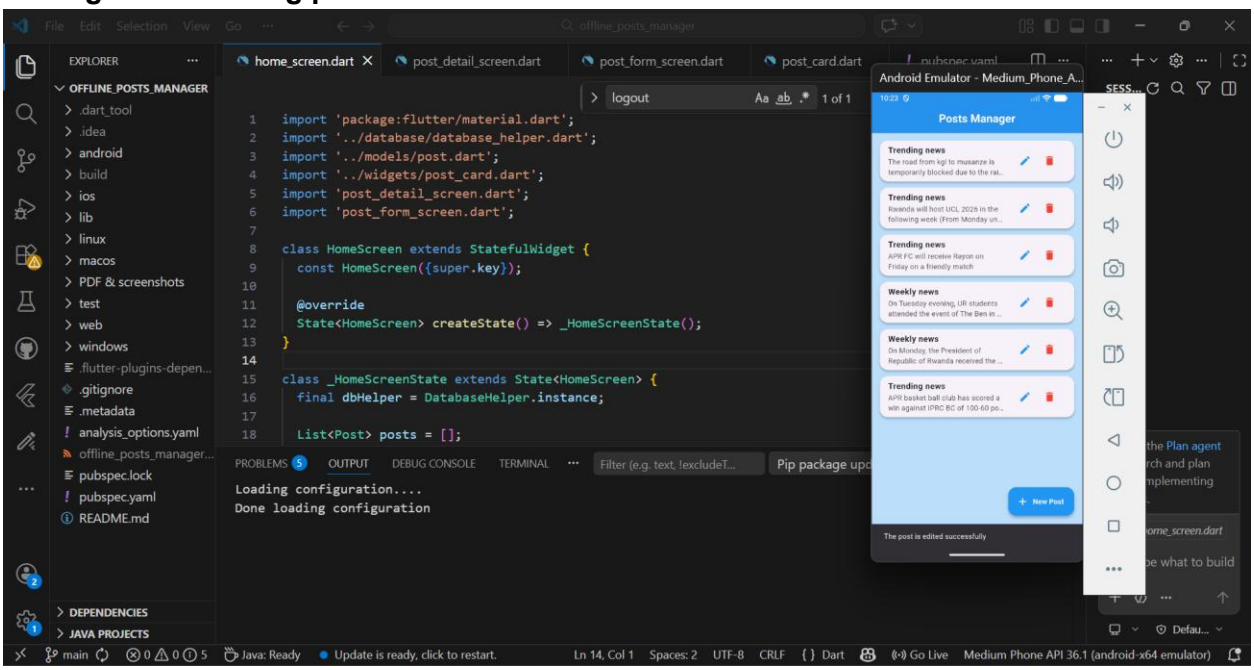


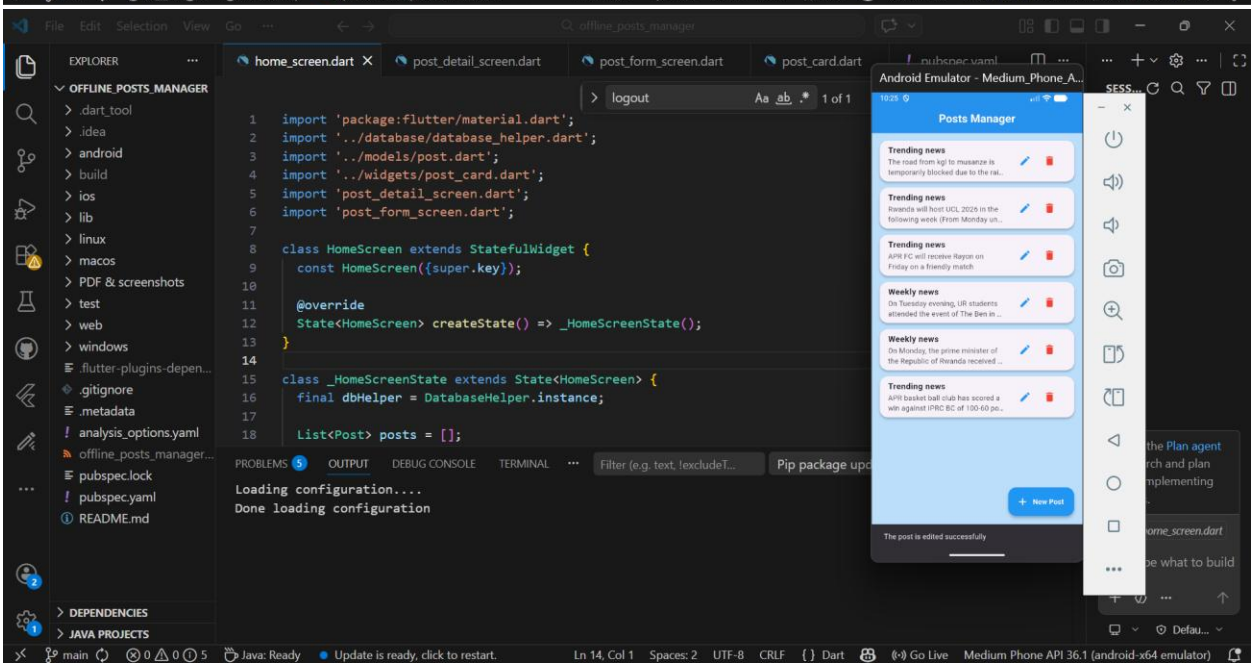
7. Editing an existing post



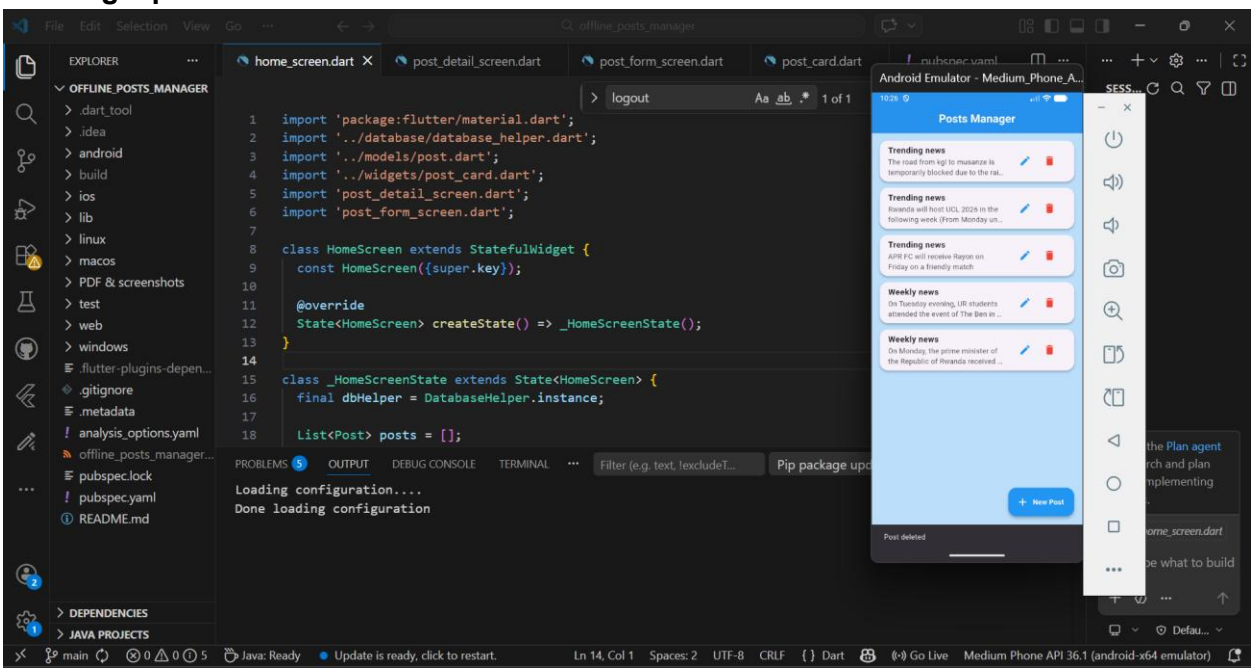


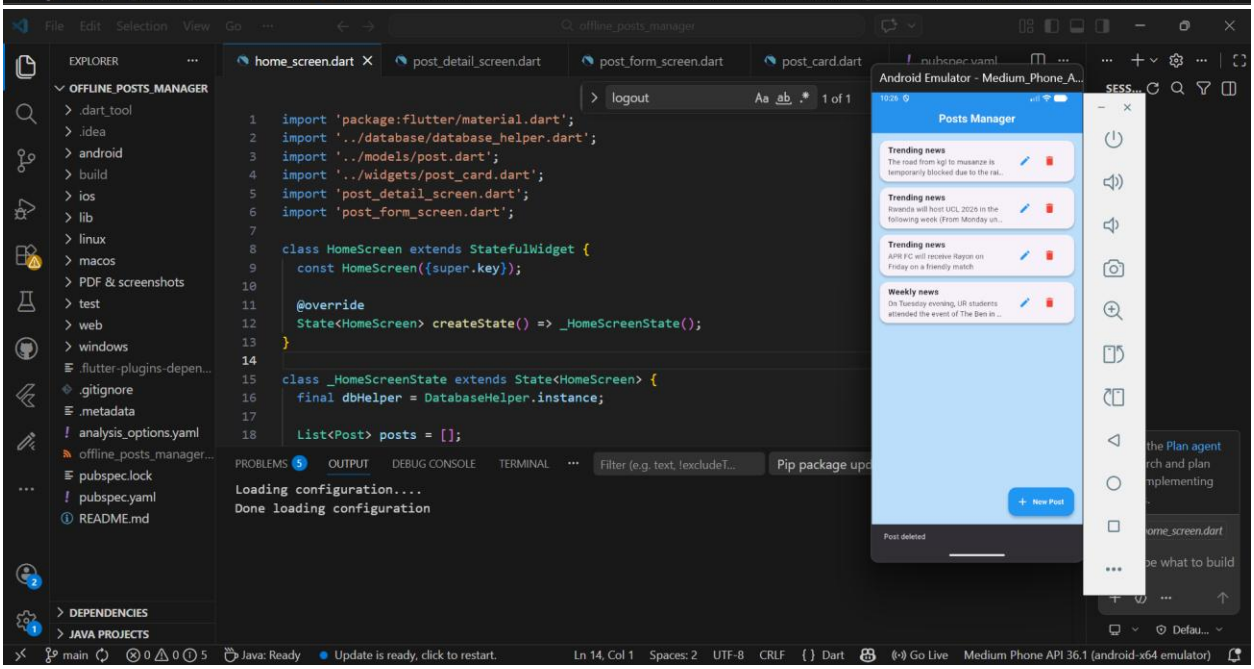
8. Editing other existing posts



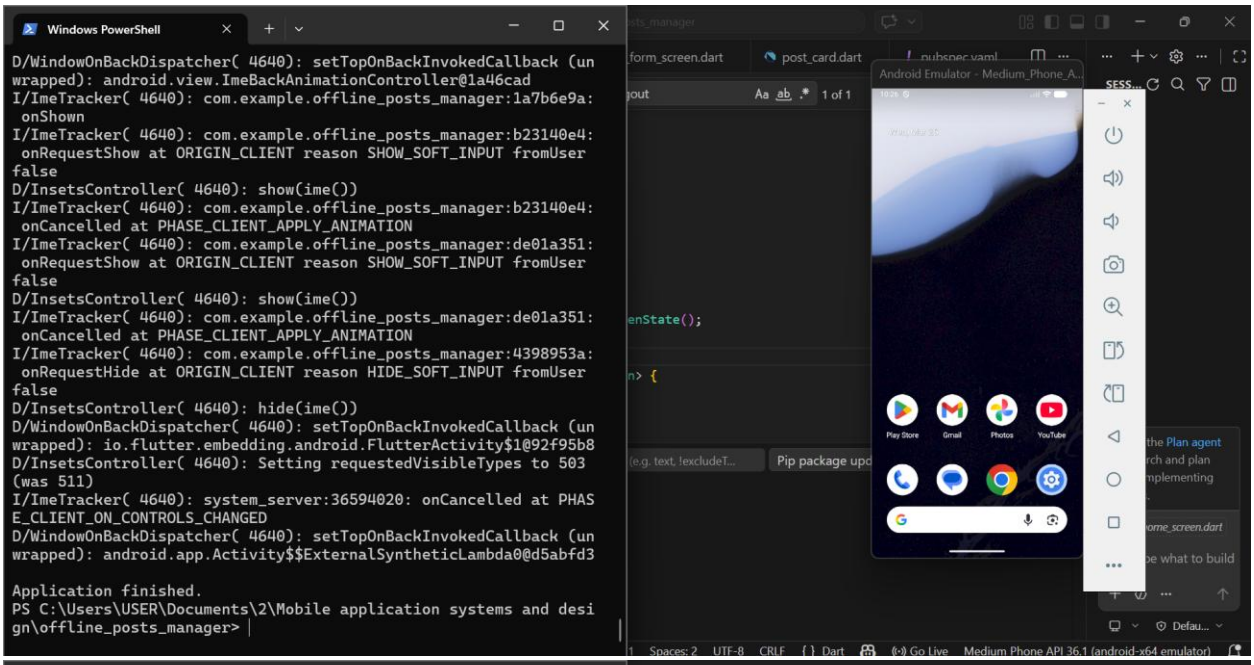


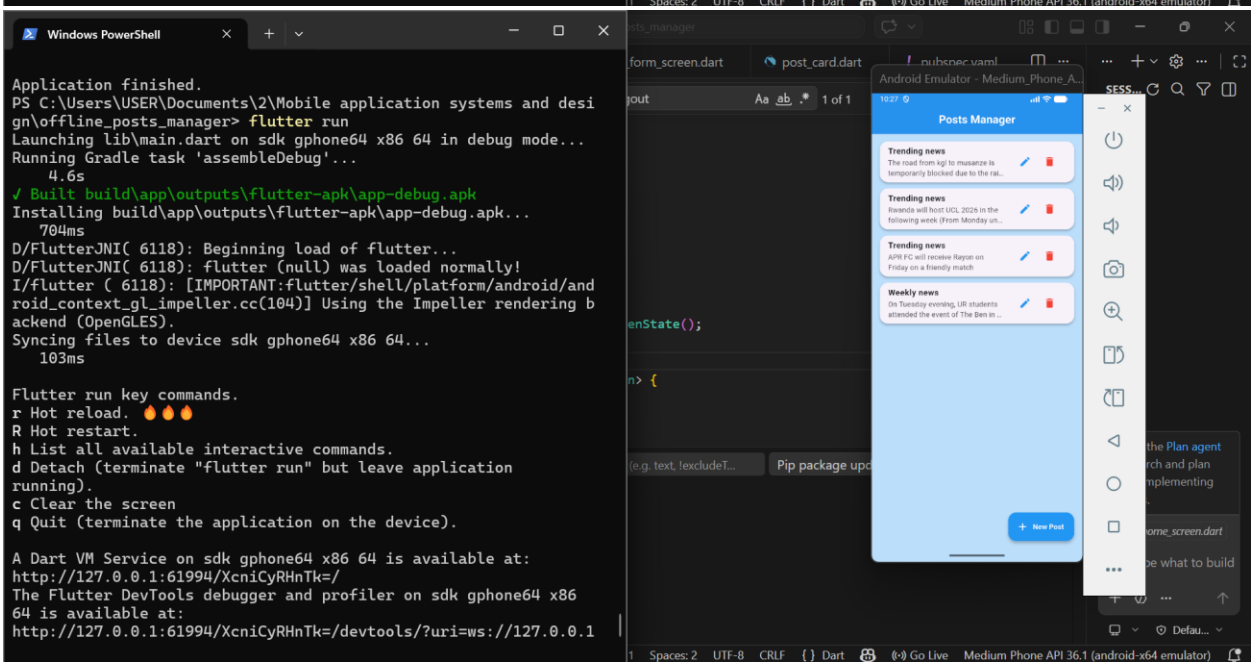
9. Deleting a post





10. The posts remain available even after the app is restarted





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